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Integration of Oculus Rift based Virtual Reality with Unity 3D in a Driving Simulator



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Preface

To get familiar with the real life situations in the domain of transportation sciences and learn about the field of work in practice, we carried out a 10 weeks internship at iNFRANEA. It is a part of our masters program at Hasselt University. The aim of this internship is to gain some practical knowledge by applying our theoretical knowledge to the real life scenarios. After the internship we are likely to have a better insight and broad visualization of the transportation field.

This report is comprised of the complete knowledge about our work on the project at iNFRANEA. The report also includes complete working of the project and factors related to the project in simple language so that it can be understandable to an average person. The distributions of the chapters are arranged such that it makes it easier for the readers to comprehend the concept of our project at the internship. We hope that we have expressed our views in this report in the most appropriate way by explaining each topic in the most detailed manner and also by not leaving any doubts in the mind of the reader.

Aim of Research

The aim of this research is to investigate about;

- Use of Unity 3D as driving simulator
- Impact of using Oculus Rift instead of screen (3X monitors and curved projection) for driving simulator

Methodology:

Based on the goals of research following methodology was used;

- Unity 3D as driving simulator
- Understanding Unity 3D feature set
- Understanding the frame work of physics library and function to make Unity 3D more suitable for driving simulation
- Integration of Oculus Rift with Unity 3D as substitute of classic driving simulator
- Eye tracking technology in Oculus Rift
- Particle Implementation of Oculus Rift as driving simulator
- Current application of Oculus Rift
- Pros and cons of using Oculus Rift – VR sickness

Expected Results:

The expected results of this research were;

- Proof of concept VR Unity 3D as driving simulator using classic environment
- Proof of concept VR Unity 3D as driving simulator using Oculus Rift

1. Organization of Internship – iNFRANEA

Infranea is a civil engineering consulting and construction firm. It is specialized in 3D designing, visualization and management of projects. They provide support from initial phase of conception to the implementation, management and maintenance phase. Being a facilitator, they integrate comprehensive approaches to contribute to a more sustainable infrastructure. For example, by finding the optimum balance between technical requirements and traffic planning, landscape needs or by reducing failure costs and disruption during implementation.



Figure 1 Logo of iNFRANEA

They work on four disciplines of a project to manage it more efficiently;

- 3D Engineering
- 3D Visualization
- Systems Engineering
- BIM – Building Information Management

Currently they have less than 50 employees in the company.



Figure 2 Working disciplines of iNFRANEA

2. Introduction

This is a group internship which consists of two main tasks:

1. Unity 3D as driving simulator
2. Integration of Oculus Rift with the SignSIM

A few years ago, iNFRANEA together with IMOB (UHasselt) and Connect developed the 'SignSIM' simulator to enable the evaluation of traffic signs and line markings in a driving simulator environment. The SignSIM is unique in its kind as it enables the integration of virtual traffic signs and markings into real-world video footage of an existing road. By this means, complete signage and markings plans (e.g. for road works) can be evaluated and tested by real road users before they are implemented. Hence, potential problems with traffic signing and marking can be detected and corrected before actual implementation on the road.

New virtual reality (VR) software such as “Unity 3D”, enable the realistic visualization of 3D-models of road infrastructure projects. Driving simulation, however, goes beyond realistic visualization in the sense that a number of relevant driving parameters (such as speed, lateral position, time-to-collision, acceleration, deceleration, etc.) need to be collected during the interaction between the user and the virtual environment, other traffic needs to be integrated in the simulation and the vehicle of the driver who interacts with the virtual environment should behave like a real vehicle (realistic vehicle dynamics). VR environments such as 'Unity' are not made in the first place to serve the purpose of driving simulation, but could be made suitable for driving simulation if the entire above are additionally programmed.

The first task of this internship therefore is to investigate which additional functional requirements for driving simulation are needed to turn 'Unity' into a driving simulation tool. In addition to this, the interaction with PTV Vissim and Unity will be researched.

Recently, new Virtual Reality (VR) technologies enable a more 'immersive' experience. For instance, the Oculus Rift is a pair of head-mounted goggles offering an unprecedented VR experience in simulated environments and could become an alternative for classic driving simulation where the driver takes place in a vehicle mockup and interacts with the virtual environment on classic computer screens. However, there are also potential downsides to more immersive VR environments, such as VR sickness (i.e. the participant may experience feelings of dizziness and sickness when interacting with the virtual environment).

The Second task within this internship is therefore to investigate the added value and potential challenges of using the Oculus Rift to interact with the SingSIM environment.

2.1 Internship Outline

The data review part for the internship consists of following steps;

- Understanding Unity 3D
 - Different feature set of Unity 3D
 - How classic driving simulation can be switched with gaming engine like Unity 3D
 - Investigate which additional functional requirements for driving simulation are needed to turn 'Unity' into a driving simulation tool.
 - How driving through existing road network is possible
 - How driving through newly designed road network is possible

- Understanding the frame work of physics library and function to make Unity 3D more suitable for driving simulation
 - How realistic the environment can be
 - Collection of important parameters like speed, lateral position, time to collision, acceleration, deceleration etc. (All important parameters which must be, should be, could be and would have to be collected to understand the behavior)
- Integrate the traffic Vissim model with Unity 3D
 - How in real time Vissim data can be integrated with Unity 3D
 - Driving behavior
 - Vehicles dynamics
- Current application of Oculus Rift
 - How realistic the experience can be
 - How to deal with VR sickness
- Pros and cons of using Oculus Rift – VR sickness
 - How to integrate eye tracking

After this part the main focus of the internship will be on developing the project and check the implementation of such project in which oculus Rift along with unity 3D used as a driving simulator.

2.2 Process Flow Diagram

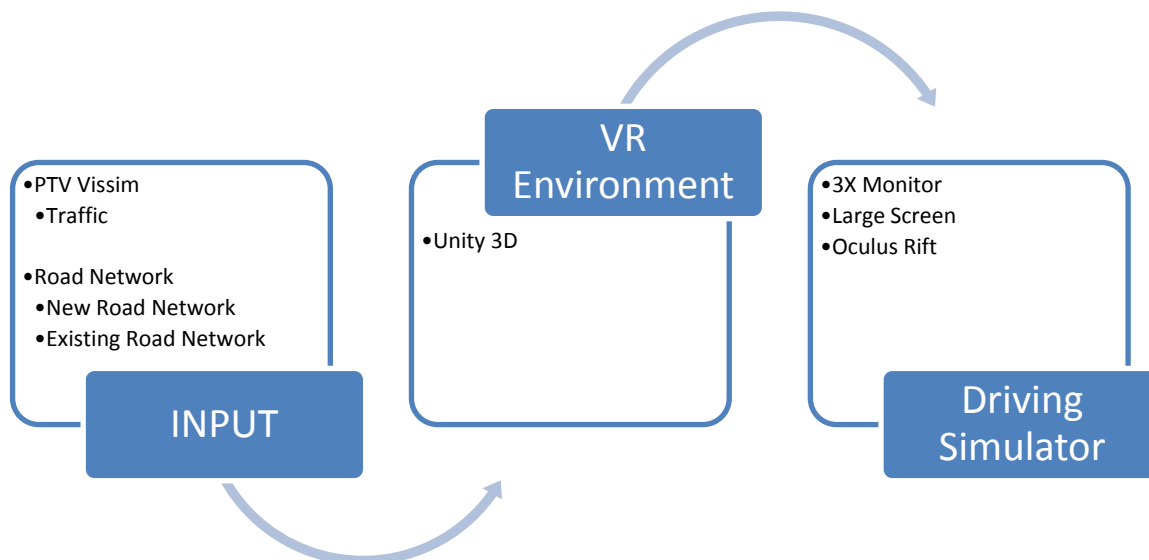


Figure 3 Project Block Diagram

3. Literature review

3.1 Virtual Environment

This chapter illustrates the definition of the Virtual Environment. It can be defined, as it is a digital place where a participant's motions are traced. The surroundings rendered or digitally generated and exhibited to the senses, relative to the motions. For instance, when a person is playing a game, the joystick movements can be controlled and, as a result, the player of the game moves ahead showing the different environment. The main goal of the virtual environment is to modify the signs with the digital cues from the original ones. The study Biocca& Levy, (1995) showed that "The blocking of sensory impressions from physical reality is a crucial part of the most compelling virtual reality (VR) experiences. The senses are immersed in the virtual world; the body is entrusted to a reality engine" (Biocca& Levy, 1995).

Virtual environment is created with the help of different hardware and software operations (Ellis, n.d.).

- Hardware Operations
- Software Operations

3.1.1 Hardware Operations to Create VE

There are three kinds of hardware operations to create the virtual environment. These are: (also shown in Fig 4).

- Sensors that are used to detect body movements of operator, for instance, head positions sensors.
- Effectors that are used to stimulate the senses of operators, for instance, stereoscopic display.
- Special purpose hardware that are used to attach effectors and sensors to establish the sensory experiences.

3.1.2 Software Operations to Create VE

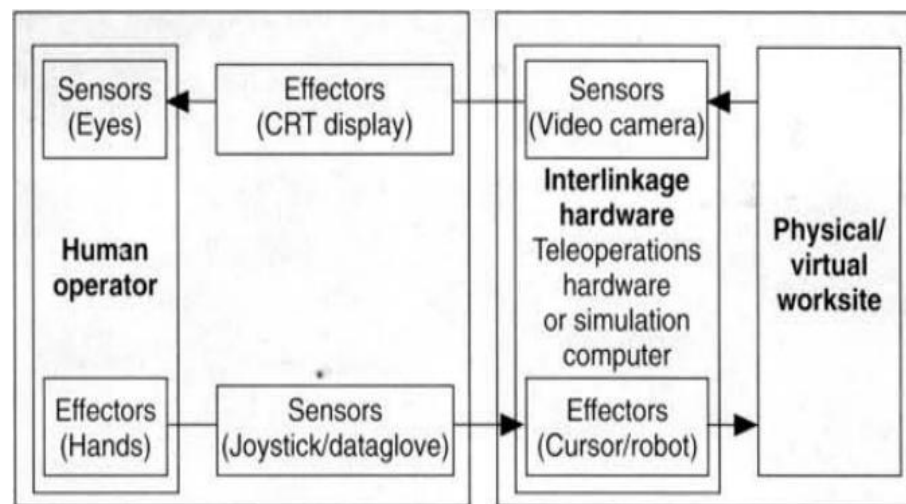


Figure 4 Functional breakdown of VE ¹

There are three kinds of software functions to create the virtual environment, which are listed as:

- Kinematics and shape of objects and actors

¹(Ellis, n.d.)

- Interactions with environment and among themselves
- Character and extent of enveloping environment.

The psychological practice of losing a person in the digital space and shutting out signs from the physical world is called immersion (Witmer & Singer, 1998). A Virtual Environment can be deployed on numerous computer-based platforms, from a screen of cellular telephone to a monitor of any desktop to a fully 3-dimensional generated virtual environment (IVE). The participant or person can move around a physical space while wearing electronic sensors that are linked to a computer. According to Figure 1 for an instance of a virtual environment, the tracing and rendering process permits a high level interactivity than traditional interactivity.

Before 1990s, the understanding of “virtual reality,” was approached with a dose of skepticism and technological fear. Now a day, the virtual reality has been an emerging technology. The truth is that most of the high-end virtual reality technology characterized and has not spread as fast as other emergent technologies (e.g., cellular phones). Since it remains expensive and unmanageable for the regularly use. In the meantime, the straighter foreword use virtual environments have been increasing.

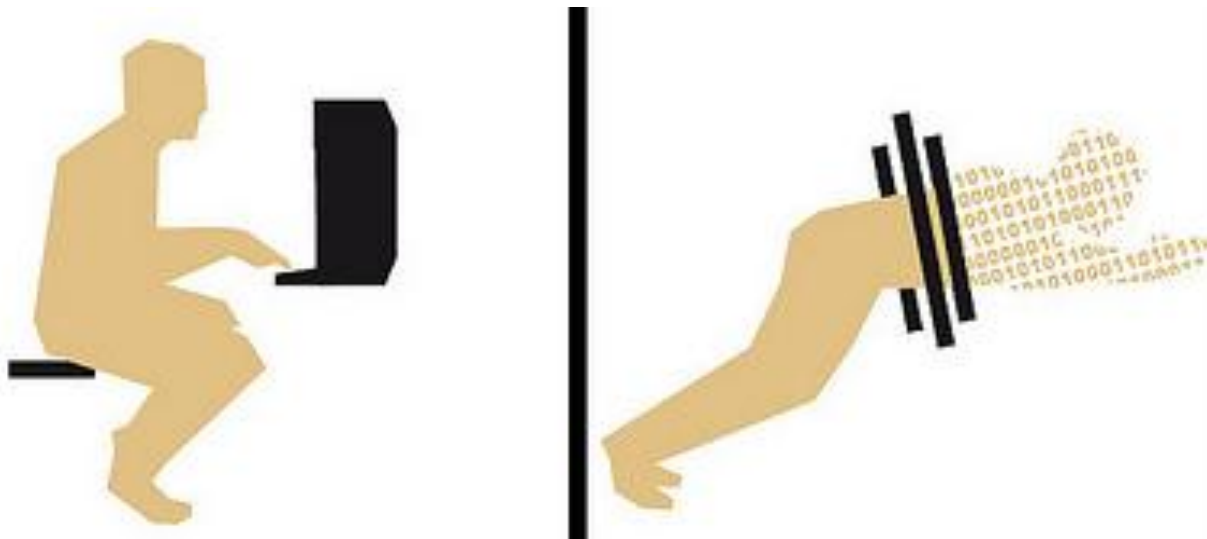


Figure 5 How VE feels

3.2 Unity 3D

Unity3D is a robust 3D engine for cross-platform development which provides a user-friendly environment for developers. It is powerful enough for professionals as well as very simple to learn for beginners. Unity3D develops interest for anyone who wants to make games in 3D and different kinds of application for desktop, mobiles, consoles and the web. (“Unity - Manual: Unity Services,” n.d.).

Following are the types of Unity 3D software along with their description:

Types of Unity 3D software

- Unity 3D
- Pro Unity 3D

Unity 3D

Unity3D is free software and has the ability to make any kind of application like smart phone applications for mobile, applications for desktop and for the web. Any development can be published on the store but the feature sets are limited in comparison to Pro Unity3D.

Pro Unity 3D

In the second type, Pro Unity3D, which requires a purchased license, has more features than Unity 3D. The user interface of Unity3D is attractive and well-structured in which the user can customize the panels according to their choice. There are different kinds of panels in Unity3D:

- **Project panel:** In this panel, all the assets are saved within the project application. They are first displayed in this panel after importing.
- **Hierarchy Panel:** In the hierarchy panel, assets are maintained in a scene. All the assets can be dragged from project panel to hierarchy panel for the purpose of adding them into the scene.
- **Inspector Panel:** In the inspector panel, all the properties and attributes of selected assets are adjusted.
- **Scene Panel:** In the scene panel, the assets are organized by dragging them around in the 3D environment.

In the project of Unity3D, assets are any materials that are used by any project or by your game. Assets include material, audio, textures, fonts, scripts and 3D models. But some objects like spheres and cubes, cannot be made directly by Unity3D. However, they can be developed with the help of other 3D modeling tools or software and then imported into the unity projects.

Scenes are very important part of the unity 3D environment where hierarchy panel shows current scene in format in a tree-like. In 3D space, the scene panel is the best or most suitable choice for managing and adjusting assets of scene. When you start working on a new application or project of unity 3D, the engine makes a new scene for you automatically. The developer spends most of the time using the hierarchy panel to arrange the scenes.

Scripts are used in Unity3D as behaviors. They allow the developer to add the assets to the project and make assets interactive. Numerous scripts may be linked with a single object and can be reused for reuse coding. Three kinds of languages can be used in Unity3D like Boo, Unity Script and C#. Unity Script is more like Action Script language and Java Script language; the second language, which can be used in Unity3D environment, is Python language. The third language, which is used and supported in the Unity3D environment, is C #language, is an object oriented language and is similar to Java. The choice of language for particular projects is dependent upon the developer skills, convenience and project requirements.

3.2.1 Interface of Unity 3D

When a user creates a new project in Unity 3D, it looks similar to the Fig 2 below. This is the basic user interface of any Unity 3D project. It has different components or panels that are used by the user to make a user interface for the Unity 3D.

1. Scene
2. Game
3. Hierarchy
4. Project/Assets View

5. Inspector
6. Graphical Icons
7. Playback Bar

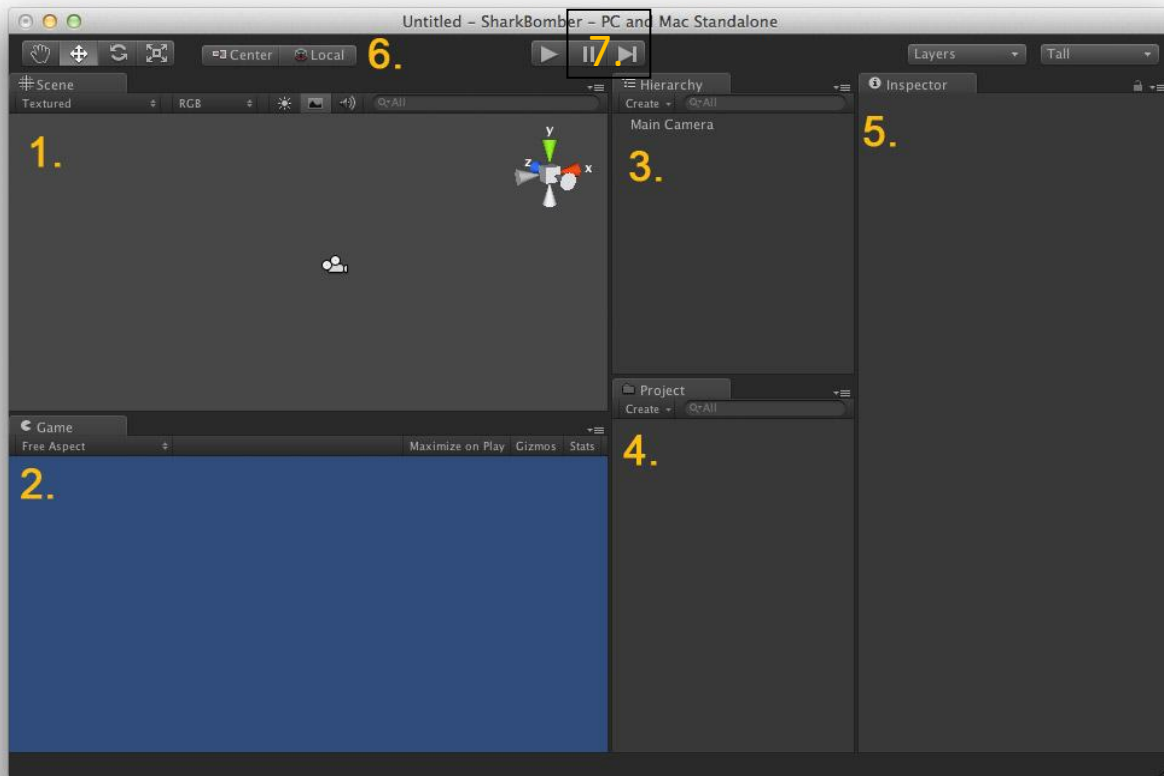


Figure 6 Basic interface of Unity 3D

3.3 Unity Libraries

Unity libraries or Unity framework are made to assist the unity developer to make innovative and entertaining games/applications/projects for the different purposes. There are numerous unity libraries or framework and each of them are used according to the needs of unity project. (“Unity - Manual: Unity Manual,” n.d.) It divides the Unity 3D model into different sections or different libraries.

Currently, the developer of video games or simulator, use physics libraries or framework to simulate actual behaviors of life in the games. This is indeed a software that gives built-in simulation which is available in Unity 3D to use. Well known physics engines are NVIDIA PhysX (“PhysX | GeForce,” n.d.) and Havok Physics (“Havok,” n.d.).

Unity has many libraries through which a unity developer can develop interactive games with ease. Following are the important unity libraries

Unity 2D

In the 2D environment, graphic objects are referred as sprites. Sprites are same like standard textures necessarily but during the development process, some important methods are used to improve the efficiency for managing and merging sprite textures. To handle the 2D physics, unity offers a standalone physics engine.

Graphics

Indeed, graphics includes the ambience, visual fidelity and rendering power. Many other features, which can be used in the development of unity 3D project, are lighting, visual effects, particles, shaders, textures, cameras and materials.

Physics

With the help of this unity library, simulation is handled and controlled easily in the unity 3D environment. Generally, there are two engines related to the physics; 3D physics and 2D physics. Both engines are the same except that the 3D dimension of 3D physics is implemented through different modules. For instance, Rigid body 2D is used for the 2D physics and Rigid body module is used for the 3D physics engine. Physics library of Unity 3D addresses the four main components are listed below:

- Rigid bodies: It is the core component, which is used to provide the physical behavior of unity 3D object.
- Colliders: It is the component of Unity 3D which describes the shape of Unity 3D object.
- Joints: This component is used to attach two rigid body objects together or one rigid body object with a fixed point.
- Character controller: It is the component which is used to provide the character a capsule shaped, simple collider.

Unity physics library and scripting in discussed in detail in section 5.3.

Scripting

In all the projects of unity 3D, scripts play very important role. These are used to handle the objects' behavior, generate the effects in terms of graphics or apply artificial intelligence behavior for objects in the unity 3D game. Indeed, in this phase, actions or events are triggered with the help of code by the developer. Different kinds of scripting tools are supported by the Unity 3D and listed below:

- Console Window
- Mono Develop
- Log Files
- Editor Test Runner
- IL2CPP

With the required settings and adjustments, the developer can create an object as if it is real or life like.

Multiplayer and Networking

This library of the unity is made for two types of user in terms of networking properties. Those developers who want to make their multiplayer project in unity 3D, they must use High Level API or Network Manager. On the other side, those developers who want to make network infrastructure, they should initiate with the Network Transport API.

Audio

This library makes the unity projects more attractive and interesting for the users. It includes real time mastering and mixing, complete 3D sound, predefined effects and snapshots. It cannot be said that a unity project is completed without having audio effects. For instance, background music in the game or sound effect with different functions.

Animation

This feature of animation contains re-targetable animations, fully controlled at processing time, call of events within the playback, appropriate and suitable transitions and hierarchies of state, merged forms for animations and many more. It makes the unity 3D games or project more interested for the users.

Navigation and Pathfinder

This feature of unity enables the developer to make characters, which can be moved within the unity environment with the help of navigation meshes, which are made by the scenes. At run time, the navigation of objects is modified through the dynamic hindrances. This phase also includes the feature like NavMesh agents, off-mesh links and NaveMesh hindrances.

Virtual Reality

Virtual reality is integrated with the unity 3D through external plugins. In the Unity 3D, object of a person is made and its movements are controlled through the code. Object behaves like it is in the real life. The same example of person, objects of cars and landscape are made and they look exactly the same as they are in real life. With the help of different features of Unity 3D, sounds affects and color schemes are integrated to make these objects more close to the real life. However, it has some disadvantages listed below.

- Every Virtual reality hardware device contains a different plugin.
- It is possible that a conflict may occur between plugins.
- After loading new version of Virtual reality Software development compatibility issues with previous games might be occur.

3.4 Unity with Other Engines

Table 1 Compression of Unity 3D with other gaming engines

Serial number	Feature	Unreal Engine 4	Unity 3D	Notes
1	Programming Language	C++	C#, Java Script	Depends upon the Developer
2	Prototypes	Fast, no line of code,	No line of code but purchase license (extension)	Same like above
3	Assets store	Good	More better, more choices	NA
4	Price	Free but you pay 5% SHARE if revenue from games >\$3000 per Quarter	Free until revenue \$100000 per year, but not complete version (complicated)	Unreal Engine
5	Cross Platforms	Supports ~6 platforms (Mainly PC+Consoles). Windows PC, Mac OS X, iOS, Android, VR, Linux, SteamOS, HTML5, Xbox One, and PS4.	Supports 21+ platforms (PC, Console, Mobile, Web) Windows PC, Mac OS X, Linux, Web Player, WebGL, VR(including Hololens), SteamOS, iOS, Android, Windows Phone 8, Tizen, Android TV and Samsung SMART TV, as well as Xbox One & 360, PS4, Playstation Vita, and Wii U	Unity
6	Documentation (Data on internet)	Normal documented	Well documented	Unity
7	Graphics	More power full than Unity	Good	Unreal Engine
8		13% of game developers use Unreal Engine 4	47% of game developers use Unity	Unity

4. Driving Simulator

The aim of a driving simulator is to provide the user a virtual environment, which replicates the real environment, so the drivers' responses and actions to particular traffic conditions can be assessed under a safe and legal environment. Driving Simulator allows the participants to communicate with the computer screen as they feel themselves, almost in a real world. The study (Gustafson & Guldbrand, 2011) presents different kinds of driving simulators, for instance, Advanced Driving Simulators, a user can feel and observe features like view angle of 360 degree and motion control (where one can feel sitting in a real car). Fixed-base driving Simulator, in which, a participant or user have a single screen in front.

Types of Driving Simulators

- Driving Simulators Using Single Monitor
- Driving Simulators Using 3X Monitor
- Driving Simulators Using Curved Screen

4.1 Driving Simulators Using 1x Monitor

In this type of driving simulator, only one screen or monitor is used with one steer and pedal. The driver can view only the front view with the help of driving simulator. It can be viewed in the Figure 7.



Figure 7 Driving Simulator using 1 Monitor

4.2 Driving Simulators Using 3X Monitors or Multiple Screens

Here three monitors are used for the visual effect of driving simulators. All screens are attached together in such a way; the driver may view everything clearly from the right and left side (see Fig. 8).



Figure 8 Driving Simulators with 3 Monitors

4.3 Driving Simulators Using Curved Screen

Visual effect of driving simulators enabled with the help of curved screen that can be viewed in the Fig.9. The user can view everything easily in one curved screen.



Figure 9 Driving Simulators with Curved Screen

4.4 Usage of Driving Simulators for Different Purposes

Driving simulators are used due to different causes and different intentions to fulfill different goals of the community listed as:

- To test new cars in the vehicle industry
- To evaluate assistance systems embedded into the cars
- To test something in driving simulation that is not safe in the real situation.
- To easily gather testing data after performing the experiments.
- Experiments with simulations are less expensive than the real world experiments in the fields.
- To provide the choice of recreating the specific situation in driving scenario.

In the study presented by the (Bach et al., 2008) described the simulated driving by showing numerous identical findings compared with the driving in real world. Overall, the field experiments showed the controlled driving to be more expensive with respect to time spent on the field experiments, economy and data analysis. This proved that driving on simulator is more economical with respect to time and cost.

Simulators are used to accomplish a variety of goals and milestones. These are being used for the last 50 years as mentioned in the study (Goode et al., 2013), one type is a driving simulator e.g. the driving simulator, planes, trains, tanks, and trucks etc. are also tested instead of testing just the operations of car only. There are many varieties of driving simulators and can be made up of different hardware and software combinations. It can be different from each other in different features too. For instance, some of the simulators used with the help of wheel steering and some of them by the keyboard. In addition to this, a simulator can be a real car. Virtual display can be provided in the simulation world by the different things, for instance, it is provided with projector screen, sometime by a simple monitor and some time by the head mounted visual display.

4.5 Advantages of driving simulators

Standardization, reproducibility and controllability

Standardization, reproducibility and controllability features are achieved with the help of driving simulation. For instance, behavior of road design, weather conditions and virtual traffic are analyzed as an operation of research aims and training needs. The study presented in (Wassink et al., 2006) explained software structure for the purpose of producing dynamic situations in a driving simulation. For the purpose of increasing the impact of training, the implemented metaphor of movie *The True man Show* released in 1998. Each and everything around the drivers reacts as driver's behavior. With the help of driving simulators, participants can drive under the same scenarios in distinct physical places. This thing is helpful to produce standardization tests and reproducible for the research findings too. On the other hand, the traffic in the real situation is largely dynamic.

Ease of data collection.

With the help of driving simulator, efficiency and accuracy are achieved easily. But in the real environment, it is difficult to get the efficient and accurate data. For instance, in the real world environment, to find the distance between a stop line and a vehicle is impossible but the study presented by (Godley et al. 2002), achieved with the help simulator.

Possibility of encountering dangerous driving conditions without being physically at risk

(Hoeschen et al., 2001) described in their study, driving simulators are useful to train the people in unpredictable circumstances as risky driving or collision avoidance practices can be dangerous on the road. In the study (Kaber et al., 2012), authors explained simulators are the means, which provide the way, where one can learn by doing the mistakes in forgiving conditions. So, with the help of driving simulators, a person can easily encounter risky driving conditions without being physically in a real world, which is safer, secure and protected for every road users.

Novel opportunity for feedback and instruction

Driving simulators provide the choice for instructions and feedback that is difficult to accomplish in physical existing vehicles. In the study (M. Lebram et al., 2006), authors stated the achievability to replay, freeze or even reset the whole scenarios.

4.6 Disadvantages of driving simulators

The driving simulators also have some challenges and disadvantages and are described in the following sections.

Limited physical, perceptual, and behavioral fidelity

The simulators with low fidelity feature generate some unrealistic results in the terms of driving behavior and at the end, they resulted inappropriate research findings. When the user choice is affected in the simulation process, which are known as Low-fidelity simulators. Due to the reason of low-fidelity simulators, users have been demotivated and then they inclined towards the real vehicles instead of depending on the simulation.

Simulator sickness

The factor of sickness in simulation is observed especially under the demanding scenarios or like the older people. The symptoms of simulators sickness can influence usability of simulators' negatively. It is a serious debate but luckily, fruitful procedural and technological guidelines are present to eliminate it according to the study (Kalasinki E.M., 1995). Simulator sickness observed less in the young drivers than the older according to the study of (Brooks et al., 2010).

5. Unity 3D Driving Simulator

This chapter illustrates the integration of unity 3d with a driving simulator. The feature set available in Unity to make a driving simulator.

5.1 Creating a Driving Simulator

A Driving simulation scenario can be created by seating in a car mock-up, using car like controls e.g. speed pedals, steering wheel, clutch, gear shifts and breaks etc. All actions of driver performing test and positions of the car mock-up are simulated in the background. Because of the safety issues, the transportation research because of safety cannot be done in real world. As described above a driving simulator consists of a car model with driver's seat, steering wheel, driving commands and simulated dynamic traffic and roadway environment around. A good driving simulator gives the driver visual, auditory, and motion experience that closely matches the real world. It includes day and night driving scenarios, vegetation, road signs, pavement markings, and traffic control devices. It collects data on vehicle speed and position, orientation, lane changing, vehicle controls and traffic signal states (Martin, Zlatkovic, & Tasic, 2012).



Figure 10A Typical Driving Simulator Scenarios

5.2 Integration of Unity 3D with Driving Simulator

A realistic driving simulator must give visual, auditory and motion effects. Furthermore, it must be able to collect data such as, vehicle position, its orientation, lane keeping and vehicle controls(Williams, Chou, Wallick, & Moyer, 2005). By using different unity 3D parameter such realistic driving simulator can be created.

View control

It has an ability to run the setup on one window (large screen) or many view windows (3X monitor display). The front, side and back views by the drivers perspective is displaced on the view window.

Input and Output system

Input of acceleration, braking and steering can be taken from the vehicle steering wheel and pedals. This information is use to update the vehicles position, attitude, speed. Based on the inputs the output of visual (view window) and audio (sound) effects are adjusted.

Audio

A driving simulator is not complete without some kind of audio, be it background music or sound effects. Unity provides flexible and powerful sound system. For 3d scenarios sophisticated features with additional features like echo and filtering is also available. It also has tool to import most standard audio file formats for playing sounds. For research purposes it can also record audio from microphone (if available) during simulation. The most common effects unity provides for driving simulator are engine noises, tires, wind, tire squeal. The audio output is the real time feedback from the input of accelerator pedal, engine RPM, tire slip and speed.

Terrain

The Terrain system allows adding vast landscapes in the scenario. At runtime, terrain rendering is highly optimized for rendering efficiency. Initially, the landscape is a flat plane but by using different tools of height, texture, grass and other details, a desired landscape can be created. Wind zone is an additional feature in terrain. When a tree is added in wind zone components, it starts bending in a realistic animated fashion.

Lights

Unity 3d supports daytime and nightlight shading. It determines the shading of an object and the shadow it casts.

5.3 Advantage of using unity 3D

The animation in unity allows full control of animation weights at runtime. It also has tremendous rendering power, visual fidelity and ambience. It has a wide range of lighting effects. The rendering affects starts from a sunny day to glowing neon sign at night. This is done by tools of lighting, cameras, materials, textures and visual effects. It allows c# and java script to write code. It is equipped with advance automated path finding and navigation meshes which allow other vehicle and character in the simulation to move along the path.

Unity allows developing a multi-player driving simulator. It is an important feature in transportation research. It allows multiple users to drive at same time on different driving simulators, on same scenario

whereas; in conventional driving simulator 1 person drives at a time. The data collected from this will be more realistic and efficient.



Figure 11 Multiplayer scenario in Unity 3D

Building a driving simulator in unity allows correct acceleration of objects and their effect of collisions, gravity and other forces. By setting few parameters a vehicle can behave passively in a realistic way. These parameters can be wheel, terrain, sphere, box, mesh and capsule collider, character controller, character, fixed, hinge, spring and configurable joint, constant force, rigid body, physic material, Joint and ragdoll stability. By changing values of these parameters the static and dynamic behavior of the vehicle will become more realistic.

5.3.1 Car Physics:

This section is about simulation of cars in unity 3D, in other words vehicle physics. How the physics libraries can be used by scripting. The vehicle physics can be illustrated in simple words when longitudinal and lateral forces are discussed separately.

Longitudinal Forces: They operate in the direction and also in opposite direction of the car body. For instance

- Wheel force
- Breaking force

- Rolling resistance
- Drag

These forces are used to control acceleration, deceleration and speed of the car.

Lateral Forces: They allow car to turn. They caused by side way friction on the wheel. These forces cause angular momentum and torque of the car.

Traction: This is the force (F_{traction}) where engine turns the wheel forward. The wheels push back the road surface, in reaction, the road surface push back vehicle in forward direction. In the Unity 3d, this force can be calculated as

$$F_{\text{traction}} = U * \text{EngineForce}$$

Where U = unit vector in the direction of car movement.

EngineForce = Tractive force controlled by the user

But in real life, the car does not accelerate to infinite speed. In real life, resistance forces are also present. These forces can be air resistance, aerodynamic drag and etc. for the driving simulator, these forces are very important because according to the mathematical expressions air resistance is proportional to the square of the velocity. When we are driving fast, this becomes the important resistance force (F_{drag}).

$$F_{\text{drag}} = -C_{\text{drag}} * V * |V|$$

Where C_{drag} = constant for the surface

V = the velocity vector

$|V|$ = Magnitude of vector V

The rolling resistance is also present in the environment. It is caused by friction between the wheel and road surface. At low speed, rolling resistance (F_{rr}) is the main resistance force. But at high speed, drag is the main component of residence force.

$$F_{rr} = -C_{rr} * V$$

Where C_{rr} = constant for the surface

For making the car more realistic F_{drag} can be rewritten as

$$F_{\text{drag}} = 0.5 * C_d * A * \rho * V^2$$

Where C_d = Coefficient of friction

A = frontal area of the car

ρ = Density of air

V = speed of the car

Breaking: a simple breaking model can be defined as

$$F_{\text{Breaking}} = -U * C_{\text{Braking}}$$

5.3.2 Integration with other hardware

Finally, Virtual reality is growing field and unity allows integration of virtual reality hardware.

6. Adding virtual reality to driving simulator

This chapter starts with the brief description of virtual reality. The history of head mounted devices (HMD) starts from 1965. Virtual reality previously and still now is mostly used for medical purposes (Carlozzi, Gade, Rizzo, & Tulsy, 2013; Xu, Chen, Lin, & Radwin, 2015). These head mounted goggles reappeared in the gaming industry in 2012 when Oculus Rift was developed (Sherman & Craig, 2002). When comparing virtual reality with fixed screen (wide screen or 3X monitors) the advantage of using virtual reality in driving simulator is that performance of driver is easy to measure in a virtual environment since the position, velocity, acceleration, and heading of the driver's vehicle is known at all times. The size of the simulator is reduced, as it only consist of steering wheel and pedals. With the help of head tracking tool driver can experience a around as well which is helpful in experiencing lane change, parallel parking, as it is not possible in fixed driving simulator. When HMD is used the driver can only see the virtual environment and senses his presence in a real driving scenario, whereas in fixed simulator a driver can see thing where the simulator resides. The simulator can compute and display, in real time, a driver's longitudinal and lateral control performance (Mourant & Schultheis, 2001).



Figure 12 Old image of Head Mounted Goggles

Driving assessment is not possible in real life scenario because of the safety, ecological and financial issues therefore these limitations are covered by virtual reality by controlled environment which is safe and financially affordable. It can also record and measure complex behaviors as well (Schultheis & Mourant, 2001).

Currently there are different virtual devices available in market. The table is provided which compares different attributes of these devices. As 5 means very good and 1 is for poor.

Table 2 Comparing different VR devices²

Device	Oculus DK2	Sony Morpheus	Samsung Gear VR	Vuzix Wrap 1200DX-VR	Vrvana	GameFace	Cmoar	Sulon Cortex	InfinittEye	Antvr	Vrelia VReye
Display Resolution	5	5	5	4	5	5	4	4	5	5	5
Display Refreshness	4	4	4	4	4	4	4	4	4	4	4
Display Type (OLED, LCD)	5	3	5	3	5	3	4	4	4	4	4
Viewing Optics (FOV)	3	3	3	2	3	3	3	3	5	3	4
Interface HDMI	5	5	1	5	5	5	1	1	5	5	5
Interface USB	5	5	4	5	5	5	4	4	5	1	1
Internal Tracking Sensor	5	4	5	5	5	5	3	5	5	5	3
Position Tracking Sensor	4	4	5	5	5	5	4	5	5	5	4
Engine Integration tools (Unre	5	2	1	1	5	4	3	1	1	1	4
SDK	5	2	1	5	4	5	1	3	2	4	4
3D Immersive Quality	4	4	4	4	4	4	4	4	4	4	4
Active User Community	5	3	4	4	3	4	3	3	3	3	3
Latency	5	4	3	4	4	5	3	3	3	3	3
Persistence	4	3	4	3	4	4	3	3	3	3	3
Stereoscopic 3D	4	4	4	4	4	4	4	4	4	4	4
Weight	3	3	3	5	3	3	3	3	4	4	3
	88,8%	72,5%	70,0%	78,8%	85,0%	85,0%	63,8%	67,5%	77,5%	72,5%	72,5%

Based on this data oculus rift is recommended and is further discussed in the next section.

²(Ihemedu-Steinke, Sirim, Erbach, Halady, & Meixner, 2015)

6.1 Oculus Rift

A brief description of Oculus rift can be given as it is light weight head mounted goggle which allows users to step into 3D at any direction. It has 3 pair of lenses. The first pair focuses on infinity whereas the rest of two pair is for the people have problem of near sightedness (Desai, Desai, Ajmera, & Mehta, 2014). Because of the structure of the oculus rift glasses can also be used along with the rift if the glasses are not long.



Figure 13 Front and Back of Oculus Rift

Unlike other VR devices Oculus rifts (OR) claims to solve the problem of motion sickness and dizziness. This is because of the head tracking technology present in Oculus Rift. The head tracking works continuously and monitors the position of head. Instead on rely on other device to change the view; the view is control by the output of head tracker which gives user a natural feel as driving in real. Because of the fast processor, the delay is between head movement and view movement is removed. This delay is the causing factor of motion sickness hence the claims seems to be valid.

Other than head tracking OR also has position tracking. The position tracking reduces the dizziness by translating the physical motion of the upper body in the real world. The working of Oculus rift is illustrated in the figure below.

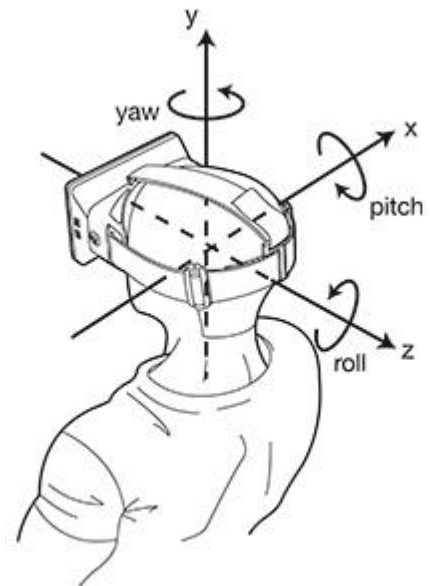


Figure 14 Head Tracking in OR

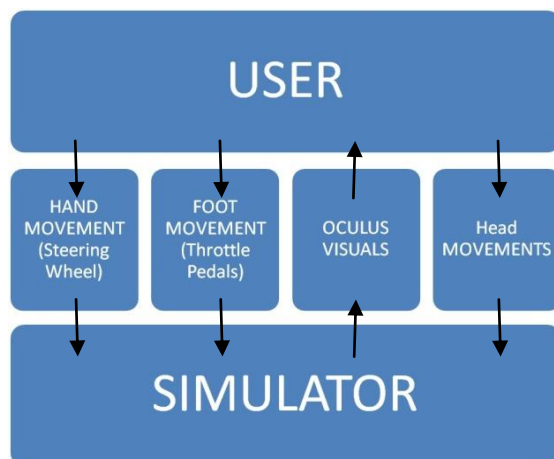


Figure 15 Basic Working of Oculus Rift

This working principle of Oculus Rift helps in combating the discomfort which arises by using simulator. This discomfort is known as Simulator Sickness ("Oculus Rift Best Practices," 2015). This discomfort is mainly occurs when the visuals from the simulator gives self motion in the absence of actual movement. Still there is no particular cause of simulator sickness because different people have different

experience and long exposure of simulator can be make the brain less sensitive to the sickness effects. Therefore the potential contributors of the simulator sickness can be; speed of movement and acceleration, degree of camera control, duration, altitude, field of view, latency, distortion, flicker and experience. Independent Visual Background is a method in literature that with the help of virtual reality can reduce the simulator sickness significantly. In this method the background is stationary but the foreground is moving around the driver.

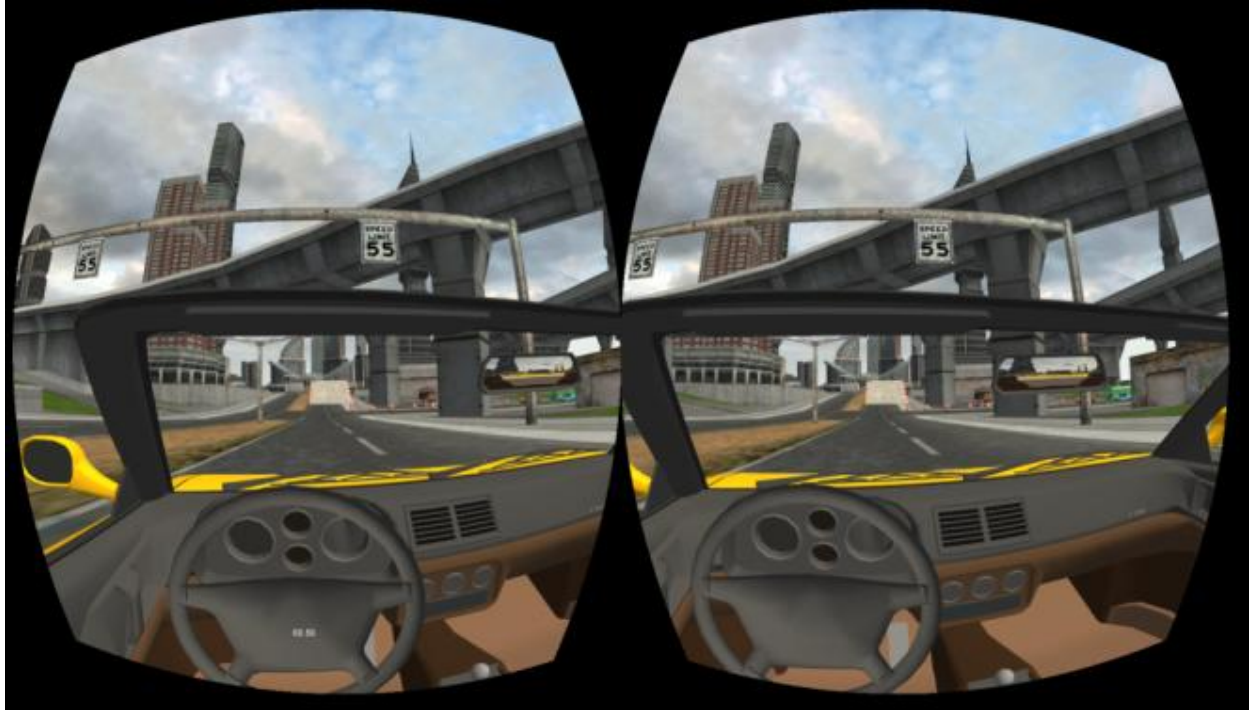


Figure 16Virtual Reality based Driving Simulator Scenario used in research (Reiners et al., 2014)

As shown in above figure a VR headset is feed with two displays, one per eye. The lenses in the headset focus and reshape the image for each eye. These lenses can be adjusted to match the distance between the driver's eyes. Due to this adjustment a 3D stereoscopic image is created by angling the 2D images provided.

7. Oculus Rift based, Unity 3D Designed Driving Simulator

To design a driving simulator in Unity 3D with the integration of Oculus Rift the hardware required can be;

- Oculus rift
- Computer – Unity 3D
- Driving simulator kit

7.1 Creating Prototype Driving Simulator

The architecture of the system is very simple. A scenario can be created in Unity 3D. Unity's asset store provides various types of inbuilt models which can be easily put into the scenario. Because of the excellent graphics, these models are very close to the real world. From the asset store vehicle physics library is also available which provides realistic vehicle controls. From the scripting tool vehicle dynamics can also be implemented. Other than this different vehicle models and sound packs are also available in the store for making the scenario more realistic and close to real world.

Resolution and the level of immersion accessible through Oculus are an important role in making the driving simulator near – realistic. Unity tools and libraries for advanced physics and wheel colliders are used to simulate a realistic car model but by object oriented scripting; car's behavior can be said close to realistic. BMW 7 series model is available in Unity 3D.

7.2 Data collection

Data collection is an important aspect of an experiment. In this through a USB cable the tracking of head is done. The speed, acceleration and steering wheel movement for lane change and aggressive and non-aggressive maneuvers is collected from the input of driving kit simulator (DSK). The input from DSK and Oculus rift controls the output display. The analytics library in Unity 3D tracks the demographics and the behavior of the driver. Different graphs, visualization tools for custom events and data explorer tool are also available in analytics dashboard.

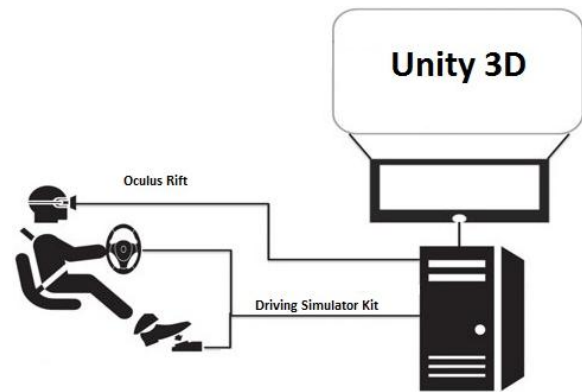


Figure 17 Architecture of Oculus Rift and Unity 3D based Driving Simulator



Figure 18 Virtual Reality based driving Simulator setup

7.3 Data

The data extracted from driving simulator consists of speed, reaction time, lateral positions, time to collision, stopping distance and etc.

One way to calculate the speed for the rigid body is by using the rigid body command in unity script.

$$rawSpeed = rigidbody.velocity.magnitude;$$

This command will give speed in meters per second. So for calculating speed in KPH this value will be multiplied 3.6.

$$KHP = rigidbody.velocity.magnitude * 3.6;$$

The speed can also be calculated for rigid body and non-rigid body object simply by taking the difference of the current frame to previous frame, the difference is then divided by the time slice to give the speed. This simple code can be written using the scripting tool of unity 3D

$$Velocity = (current - previous) / Time;$$

This speed will be displayed through speedometer. This speedometer can be created by GUIText gameobject from game object menu. By scripting the speedometer displays the speed in desired unit.

The reaction time can be calculated by the time difference using command Time.time whereas if pausing is come into play then time.deltaTime can be used. Vector direction can be calculated by subtracting one point of space from another. The magnitude of the vector is equal to the distance between the two objects position. Similarly, the stopping distance and time to collision can be calculated by physics formulas written in scripting tool of Unity 3D.

8. Results and Conclusion

To gain some practical findings of this report, a test was carried out at imob using the driving simulator. This research consists of two scenarios. First the participants tested the driving simulator using a 3d model environment that was not made in Unity. In the second scenario they tested the driving simulator using the Unity 3d model environment. At the end of the test, feedback was taken from one of the participant about the experience of testing the driving simulator (with and without unity 3D), the participant responded as;

“While testing the driving simulator without the unity 3D model, I experienced some strange feelings in my stomach which were dragging me towards simulator sickness. While in test 2 when Unity 3D model was used, I felt better and had no strange feelings.”



Figure 19 Image taken during test 2 at imob during internship

Virtual reality is tested by different transportation researchers in one research (Reiners, Wood, & Gregory, 2014), to evaluate the potential of virtual reality test is conducted in which a sample size of 13 is taken. The sample includes six males and seven female adults. The participants were tested with scenario. After the test they were ask to rate the virtual reality from the sale of 1 to 7 (1= strongly disagree, 4= Neutral, 7 = strongly agree).

The results are attached below.

Table 3 Results from the study by Reiners et al.,

Question	Mean Value	Standard Deviation
Realistic	6.8	0.44
Usable	6.8	0.38
Interesting	6.8	0.38
Engaging	6.8	0.38
Compelling	6.7	0.48

This study can be concluded as follows;

- Unity can be used as standalone driving simulator.
- Scripting supports current physics library to give acceptable results.
- For more accurate results, vehicle dynamics must be adjusted with the advanced formulas.
- If the real traffic values are used for any scenario the outcome of the research will be more realistic.

The actual testing of Unity 3D was not completed because of the time limitation, although, small test was done as discussed above. Without full testing it is difficult to say that the expected results will be achieved.

9. Recommendations

Due to project commitments the initial testing of Unity as a stand-alone driving simulator was not able to be conducted. However, for iNFRANEA to be able to proceed with the initial testing the following procedures and minimal requirements are recommended:

- Use of advance physics library, to make vehicle dynamics more realistic in simulator.
- Integration with Vissim for real traffic situations.
- Better graphics and advance rendering.
- Use of multiplayer network can lead to results that are similar to real life.

10. Future Work

The purpose of a driving simulator is to overcome the immersion experienced lack in 2D driving simulator (Ihemedu-Steinke, Sirim, Erbach, Halady, & Meixner, 2015). The level of immersion is limited to the occurrence of simulator sickness. This sickness can be due to the poor graphical performance of the driving scenes, low resolution and incorrect scaling of the speed and acceleration. When these limitations are removed, an efficient and near-realistic driving experience of hazardous driving situations without endangering the life of the driver can be made.

The final piece of VR puzzle is the eye tracking. Oculus rift does not provide the eye tracking technology directly. But it can be achieved by altering the structure of Oculus rift and installing infrared sensors monitors in the head set (Stengel, Grogorick, Eisemann, Eisemann, & Magnor, 2015). They will track where the eye is looking in virtual reality. As the system knows what the driver is looking and focusing at, due to this the graphics engine adjusts the focus and the rendering results will appear more real.

If the traffic in the model is induced from a microscopic model, then the results achieved from the virtual reality will be more real and close to real world. In future this can be done by integrating the unity model with PTV – Vissim. This integration can take place in 3 different ways;

1. Using Vissim DLL function
 - a. DLL function runs as a separate program.
 - b. When simulation start, the simulator pass position of vehicles to Vissim.
 - c. By parallel running real time transfer information from Vissim to simulator is done.
2. Vissim to 3D max to Unity 3D

- a. Create traffic simulation (FZP) file format in Vissim.
 - b. The FZP file contain information of traffic animation and vehicles data.
 - c. Use 3d max tool to open FZP file format.
 - d. Opening the 3d max file in Unity 3D.
3. Using OpenTraffic middleware
 - a. It integrate microscopic traffic simulation with Unity 3D.
 - b. For this purpose, Unity 3D multi network tool Distributed Virtual Environment (DiVE) is used.
 - c. The DiVE module updates the information of human driven vehicle
 - d. OpenTraffic middleware with the update of simulator vehicle controls autonomous vehicles.

11. Data log

Important dates

DATE	TASK
12-February	Introductory interview with 2 supervisors and signing the internship agreement
07-March	Start of internship
18-March	Submit a detailed plan of approach
25-March	Feedback on internship proposal by supervisors
26-March	Beginning of Easter holiday
10-April	End of Easter holiday
09-May	Interim evaluation (student and its 2 supervisors)
27-May	End of internship
10-June	Submit internship report (BB + email to 2 supervisors)
17-June	Upload poster
21-June	Poster presentation & final evaluation

Daily log

WEEK	DATE	DAY	TASK
1	07-MARCH	MONDAY	Introduction with Infranea employees, meeting with Angus (Supervisor), Meeting with Angus about Internship topic
	08-MARCH	TUESDAY	Meeting with Angus about Research Proposal, e.g. Design of Research proposal, Main objectives of Our Internship Report etc.
	09-MARCH	WEDNESDAY	Research and Development regarding objectives of Internship Report.
	10-MARCH	THURSDAY	First Draft of Research Proposal
2	14-MARCH	MONDAY	Meeting with Angus regarding the Research proposal draft, feedback from Angus
	15-MARCH	TUESDAY	Literature Review
	16-MARCH	WEDNESDAY	Literature Review
	17-MARCH	THURSDAY	Final Version of Research Proposal and accepted by the Angus

3	21-MARCH	MONDAY	Literature Review (Unity 3D)
	22-MARCH	TUESDAY	Literature Review + Meeting with Angus
	23-MARCH	WEDNESDAY	Literature Review + (Installing Unity 3D)
	24-MARCH	THURSDAY	Literature Review
4	11-APRIL	MONDAY	Exploring Different Unity 3D projects
	12-APRIL	TUESDAY	Preparation of testing Unity 3D Model at iMob
	13-APRIL	WEDNESDAY	Preparation of testing Unity 3D Model at iMob
	14-APRIL	THURSDAY	Execution of Unity 3D at iMob
5	18-APRIL	MONDAY	Exploring Different Unity 3D projects
	19-APRIL	TUESDAY	Exploring Different Unity 3D projects
	20-APRIL	WEDNESDAY	Literature Review (Vissim)
	21-APRIL	THURSDAY	Literature Review (Vissim)
6	25-APRIL	MONDAY	Literature Review (Oculus Rift)
	26-APRIL	TUESDAY	Literature Review (Oculus Rift)
	27-APRIL	WEDNESDAY	Literature Review (Oculus Rift)
	28-APRIL	THURSDAY	Literature Review (Oculus Rift)
7	02-MAY	MONDAY	Literature Review (3D Max)
	03-MAY	TUESDAY	Literature Review (3D Max)
	04-MAY	WEDNESDAY	Literature Review (Virtual reality)
	05-MAY	THURSDAY	Literature Review (Virtual reality)
8	09-MAY	MONDAY	Literature Review (Virtual Environment)
	10-MAY	TUESDAY	Meeting at iMob with Angus and Tom (External and Internal Supervisors)
	11-MAY	WEDNESDAY	Literature Review (Virtual Environment)
	12-MAY	THURSDAY	Literature Review (Virtual Environment)
9	16-MAY	MONDAY	Exploring Different Vissim projects

	17-MAY	TUESDAY	Exploring Different Vissim projects
	18-MAY	WEDNESDAY	Exploring Different Vissim projects
	19-MAY	THURSDAY	Exploring Different Vissim projects
10	24-MAY	MONDAY	Exploring Different Unity 3D projects
	25-MAY	TUESDAY	Exploring Different Unity 3D projects
	26-MAY	WEDNESDAY	Exploring Different Unity 3D projects
	27-MAY	THURSDAY	Exploring Different Unity 3D projects

Personal reflection

The internship project was nice, because it was a kind of new topic for us. We haven't really done any assignments about driving simulator in particular before. We started our research with some literature review on unity tools and virtual reality. We were happy to do some more practical research in the end by implementing the Unity 3D model on driving simulator.

The internship was a fantastic experience for us. We gained knowledge regarding the multi-dimensionality of the transportation field. It was a great experience for us to use our knowledge and implement a prototype on the latest state-of-the art IMOB's driving simulator. It provided us an insight into the latest research topics that are being carried out all over the world in different traffic consulting companies.

We had only the basic knowledge about the programming languages but we took the challenge as it would push us to gain more knowledge and experience in the areas not yet explored by us, which was primarily based on Unity 3D. However, with the guidance of our supervisors and dedication, we got acquainted with the Unity 3D and were able to successfully complete all the tasks assigned during the internship period. The second task of the internship regarding Oculus Rift was very cumbersome yet very interesting. It gave a practical know-how to develop a simulator in virtual reality.

Overall, the working environment at iNFRANEA was very warm and pleasant. The time spent during the internship was full of new experiences and we gained valuable knowledge not only in the technical related to our research but also in the professional environment as well. Therefore, it can be concluded that the internship not only provided us with an opportunity to familiarize ourselves with an actual working office environment but also enriched us with the practical and latest knowledge.

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